

Problem-2

Consider a silicon PN junction with an acceptor concentration $N_A=10^{17} \text{ cm}^{-3}$ in the P-type region and a donor concentration $N_D=10^{16} \text{ cm}^{-3}$ in the N-type region. The built-in potential V_0 is 0.8 V. Calculate the depletion width when a reverse bias of 5 V is applied to the junction.

Constants

- Elementary charge, $q=1.6 \times 10^{-19} \text{ C}$
- Relative permittivity of silicon, $\epsilon_r=11.7$
- The permittivity of free space, $\epsilon_0=8.854 \times 10^{-14} \text{ F/cm}$